Comsat University Islamabad
Attock campus

Name	Talha Ahmad
Reg No	Fa23-bse-025
Subject	MAD
Assignment	2
Submitted to	Sir Kamran

1. Section: Initializing the Project:

I initialized the React Native project and verified the device connection using the Android Debug Bridge (ADB). As shown in the terminal, the Pixel 5 device is successfully attached and the reverse proxy for port 8081 is configured to allow communication between the laptop and the mobile device.

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\talha>adb reverse --remove-all

C:\Users\talha>adb reverse tcp:8081 tcp:8081
8081

C:\Users\talha>adb devices
List of devices attached
Pixel 5         device
```

2. Section: App.tsx (CODE):

The App.tsx file contains the core logic of the application. It utilizes the useState hook to manage the list of tasks and the current input text. The addTask function updates the state using the spread operator, and the deleteTask function uses the .filter() method to remove items based on their index.

```
Welcome   App.tsx > App > handleAddTask > newTask
import React, { useState } from 'react';
import {
  View, Text, TextInput, TouchableOpacity,
  FlatList, StyleSheet, Keyboard, SafeAreaView
} from 'react-native';

// Define the interface for our tasks
interface Task {
  id: string;
  text: string;
}

export default function App() {
  const [task, setTask] = useState<string>('');
  const [taskList, setTaskList] = useState<Task[]>([]);

  const handleAddTask = () => {
    if (task.trim().length > 0) {
      // Create a new task object with a unique ID
      const newTask: Task = {
        id: Date.now().toString(),
        text: task
      };
      setTaskList([...taskList, newTask]);
      setTask('');
      Keyboard.dismiss();
    }
  };

  const deleteTask = (id: string) => {
    // Filter out the task with the matching ID
    const updatedList = taskList.filter(item => item.id !== id);
    setTaskList(updatedList);
  };

  return (
    <SafeAreaView style={styles.container}>
      <Text style={styles.title}>My To-Do List</Text>
```

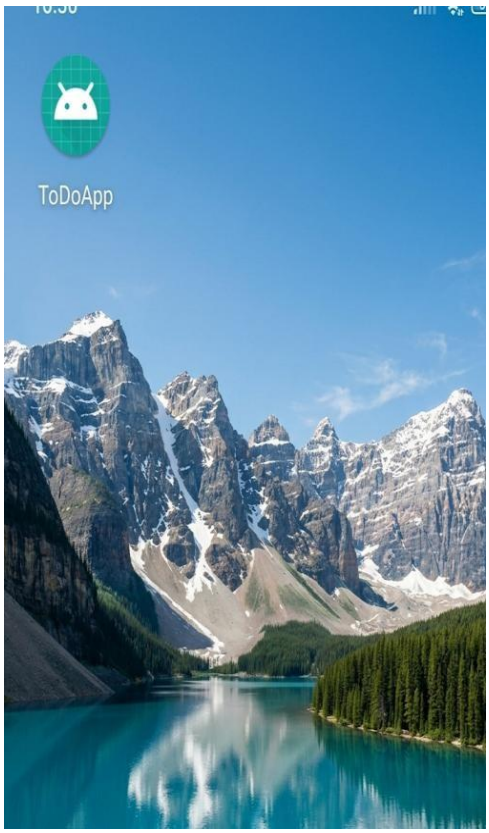
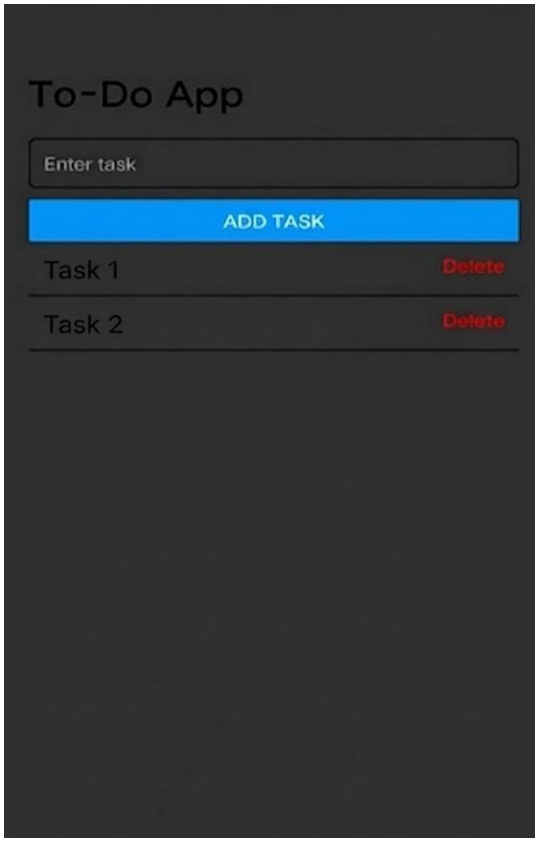
```
40          <View style={styles.inputWrapper}>
41            <TextInput
42              style={styles.input}
43              placeholder={'Write a task'}
44              placeholderTextColor="#999"
45              value={task}
46              onChangeText={text => setTask(text)}
47            />
48            <TouchableOpacity onPress={handleAddTask}>
49              <View style={styles.addWrapper}>
50                <Text style={styles.addText}>+</Text>
51              </View>
52            </TouchableOpacity>
53          </View>
54
55          <FlatList
56            data={taskList}
57            keyExtractor={(item) => item.id}
58            renderItem={({ item }) => (
59              <View style={styles.item}>
60                <Text style={styles.itemText}>{item.text}</Text>
61                <TouchableOpacity onPress={() => deleteTask(item.id)}>
62                  <Text style={styles.deleteText}>Delete</Text>
63                </TouchableOpacity>
64              </View>
65            )}
66          />
67        </SafeAreaView>
68      );
69    }
```

```
const styles = StyleSheet.create({
  container: { flex: 1, padding: 20, backgroundColor: '#1E1E1E' },
  title: { fontSize: 26, fontWeight: 'bold', color: '#FFF', textAlign: 'center', marginVertical: 20 },
  inputContainer: { marginBottom: 20 },
  input: { borderWidth: 1, borderColor: '#444', padding: 15, marginBottom: 10, borderRadius: 8, color: '#FFF', backgroundColor: '#2D2D2D' }
  taskItem: { flexDirection: 'row', justifyContent: 'space-between', padding: 15, backgroundColor: '#2D2D2D', borderRadius: 8, marginBottom: 10
  taskText: { fontSize: 16, color: '#FFF' },
  deleteButton: { color: '#FF5252', fontWeight: 'bold' },
});
```

3. Section: Project Running (Terminal)

After configuring the environment, I executed the command npx react-native run-android. The build process was successful, as indicated by the 'BUILD SUCCESSFUL' message in the terminal, and the APK was automatically installed onto the connected device.

```
C:\WINDOWS\system32\cmd.exe
C:\Users\talha\Desktop\ToDoApp>npx react-native run-android
Info A dev server is already running for this on port 8082.
The system cannot find the path specified.
Info Launching emulator...
warnor Failed to launch emulator. Reason: No emulators found as t of 'emulator -list-avds'.
Warn Please launch an emulator manually or connect.  Otherwise app may fail to launch.
Info Installing the app...
> Task :app:installDebug
Installing APK 'app-debug.apk' on 'V2061 - 12' for :debug
Installed on 1 device.
[Incubating] Problems report is available at: file:///C:/Users/C:/Users/talha/Desktop/Tald/reportss/problems-report.html
Deprecated Gradle features were used in this build, k incompatible with Gradle 10.
You can use '--warning-mode all' to show the individual deprecations and determine if they come from your own scripts or plug
For more on this, please refer to https://docs.gradle.org/9.0.ido/command_line_interface.htmlsc_line_warnings in the Gradle
BUILD SUCCESSFUL in 12s
#2 actionable tasks: 2 executed, 73 up-to-date
Consider enabling configuration cache to speed up this build: http:/gradle.org/9.0.0/userguide/configuration_cache_enabling.html
Info Connecting to the development server...
Warn Failed to connect to development server using "adb reverse": spawnSync C:\Users\talha\DelmaAndroidStudio\Sdk\platform-tools\adb ENOENT
Info Started to find the app.
Error: Command failed with exit code 1: C:\Users\talha\Desktop\AndroidStudio\Sdk\platform-tools\adb shell am start -n com.todoapp/c
-a android.enent action MAIN -n android.intent.category.LAUNCHER
    at makeError (C:\Users\talhar\Desktop\ToApp\node_modules\execa\lib\error.js:60:11)
    at module.exports.sync (C:\Users\talha\Desktop\ToDoApp\node_modules\execa\index.js:194:17)
    at tryLaunchAppOnDevice (C:\Users\alha\Desktop\ToDoApp\node_modules\@react-native-community\cli-platform-android\build\commar
hAppOnDevice.js:52:22)
    at C:\Users\talha\Desktop\ToDoApp\node_modules\@React-native-community\cli-platform-android\build\commands\runAndroid\runOnA
```

4. Section: Screenshots From Mobile Phone:

The first image shows the 'ToDoApp' icon successfully installed on the Android home screen. The second image displays the functional application interface running on the device, demonstrating the ability to add and display multiple tasks (Task 1 and Task 2) within a styled dark-themed list.